DMarlon Studio

User manual

DMarlon Studio manual

The complete guide to the visual IDE for building Python apps

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dmarlonstudio.com

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Chapter 1

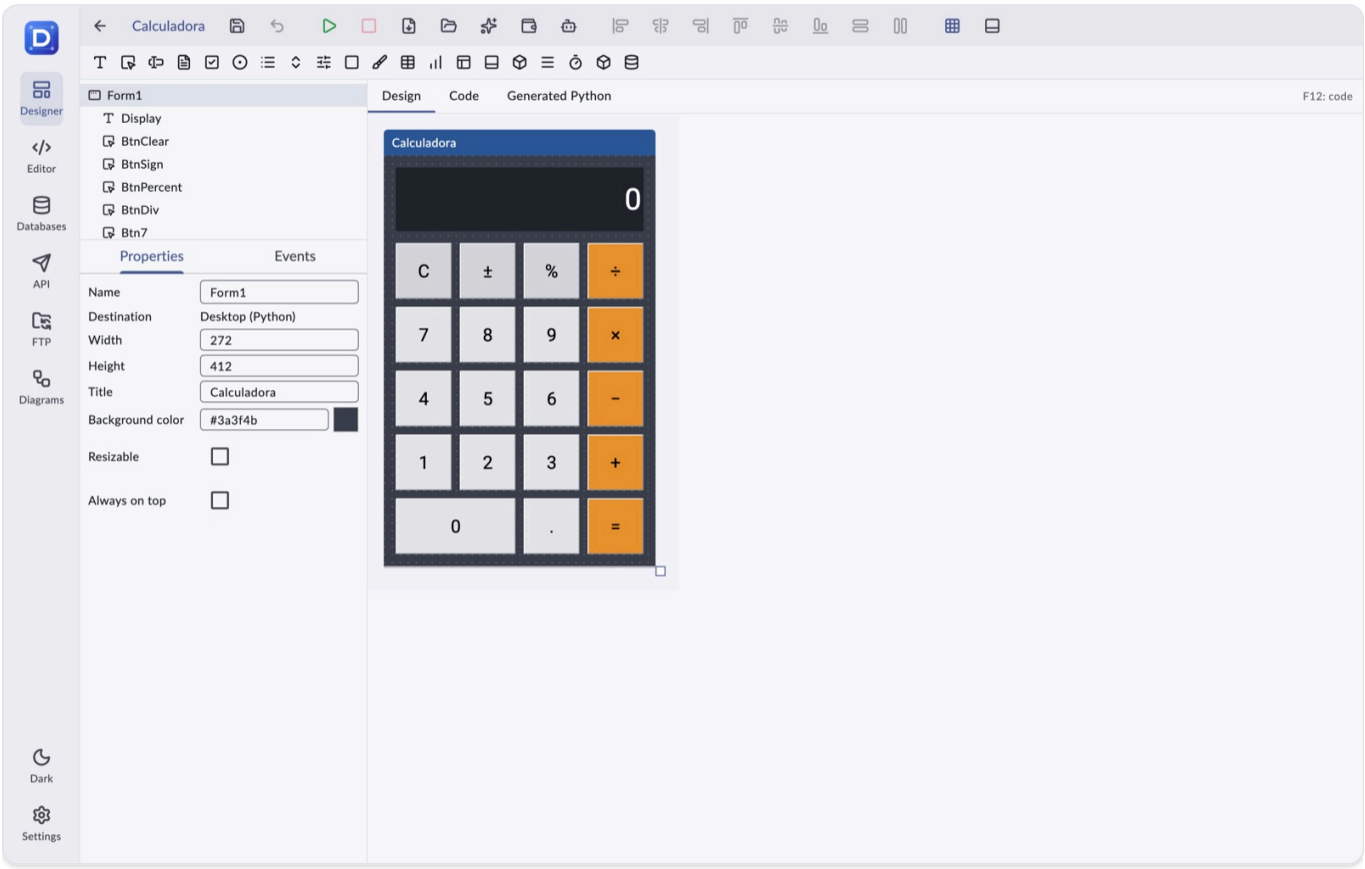
Introduction and installation

What DMarlon Studio is, what it needs and where it keeps everything.

What DMarlon Studio is

DMarlon Studio is a visual desktop IDE. You design windows by dragging components onto a form, adjust their properties in an inspector and write the code of each event in Python. **F9** runs the window on your computer.

Today the same designer builds two kinds of app: desktop windows with **tkinter** and web apps with **Flask**. Around the designer there is a VS Code-style code editor with git and a terminal, a database manager, an API client, an FTP client, a diagram editor and AI assistants (Claude by default).



The Designer: palette on top, tree and inspector on the left and the form in the center.

Requirements

To...	You need
Use the studio	macOS 10.15 or later, or 64-bit Windows 10/11.

To…	You need
Run desktop windows	Python 3 with tkinter. On macOS: `brew install python-tk` or the python.org installer. On Windows: the python.org installer with the **tcl/tk and IDLE** option ticked.
Run web apps	Flask in that same Python: `python3 -m pip install flask` (on Windows, `py -3 -m pip install flask`).
Version control in the Editor	git installed on the computer.
Claude Code chat in the Designer	Claude Code installed and signed in (optional).
AI assistants	An API key for the provider chosen in **Settings** (optional).

ⓘ The studio finds Python on its own: on the PATH, in the Homebrew folders and in the python.org install, and keeps the one with the newest Tk. The result is shown on the Designer's project list.

Installation

Download the zip for your system from the downloads section of dmarlonstudio.com.

1. **macOS**: unzip `DMarlonStudio-macOS.zip` and move **DMarlon Studio** to the Applications folder. The first time, open the app with right-click → **Open**: it is not signed.
2. **Windows**: unzip the whole `DMarlonStudio-Windows.zip` wherever you like (not in Program Files) and run `DMarlonStudio.exe`. If “Windows protected your PC” appears, click **More info** → **Run anyway**.

Where everything is stored

What	Where
Designer projects	One folder per project in `~/Documents/DMarlon Studio/Proyectos` (it can be changed in the Designer). They can be versioned with git.
Settings, connections, API requests, history and diagrams	`studio.sqlite`, in the app's data folder: `~/Library/Application Support/com.dmarlonstudio.studio` on macOS and `%APPDATA%\DMarlon Studio\DMarlon Studio` on Windows.
Deleted projects	The `.papelera` folder inside the workspace: nothing is erased for good.

Each project folder holds `project.dmsproj` (the design, as JSON), the window's code (`Form1.py`, named after the form), a `.gitignore` and, once run, `build/` with the generated Python.

Chapter 2

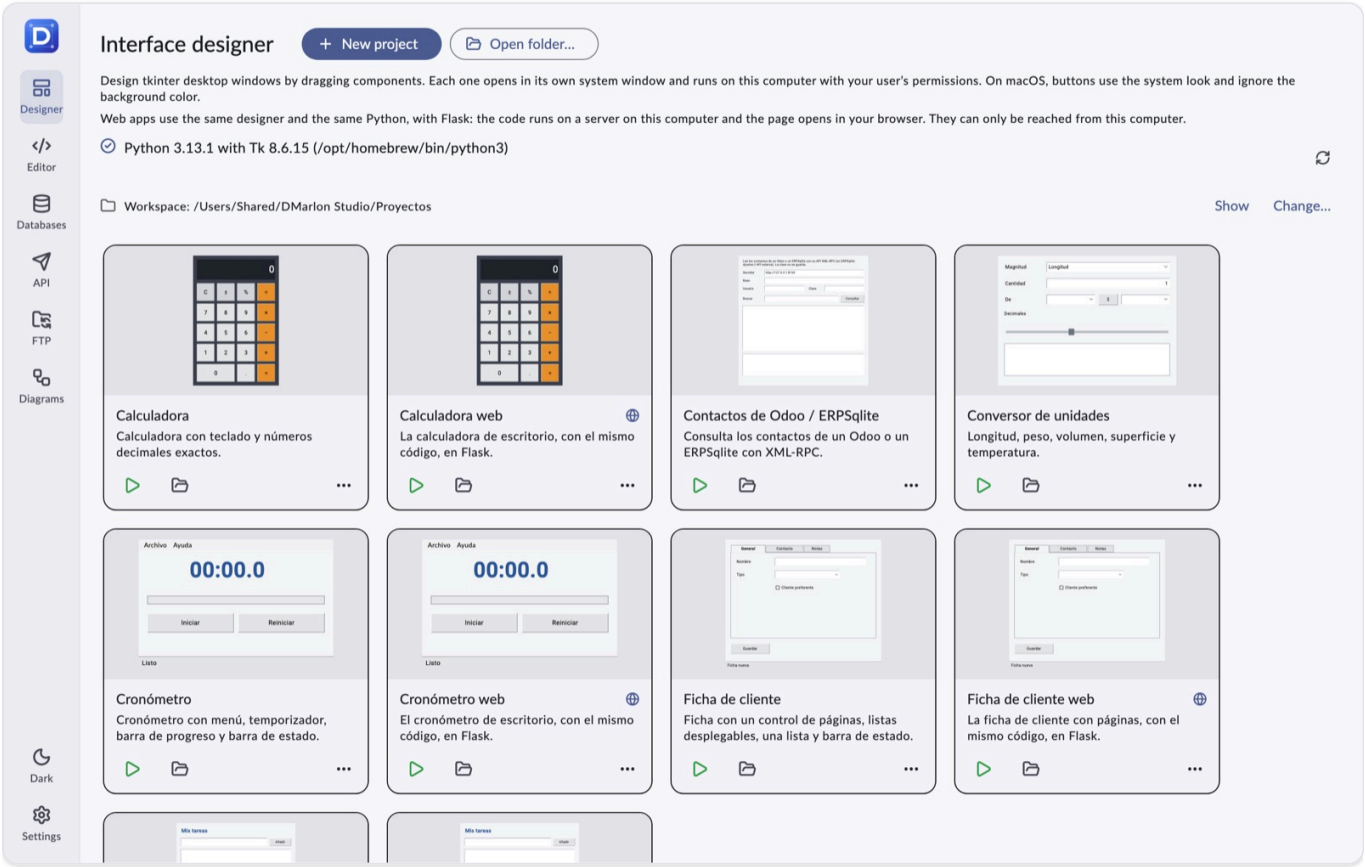
Getting started

The main window, the modules, the theme and the language.

The main window

On the left is the module bar: **Designer**, **Editor**, **Databases**, **API**, **FTP**, **Diagrams** and, at the bottom, the **Dark**/**Light** button and **Settings**.

Each module is created the first time you open it and keeps its state: you can go from the Designer to the Editor and back without losing anything unsaved.



The project list, with the examples that come with the studio.

Theme and language

- The **Dark**/**Light** button on the bar switches the theme at once. In **Settings → Appearance** you can also choose **System** to follow your computer.
- **Settings → Language** switches between **Español** and **English** without closing anything. The first time, the system language is used.

Your first window in five steps

1. In **Designer**, click **New project**, type a name and choose **Desktop (Python)**.
2. Drag a **Label**, a **Text field** and a **Button** from the palette onto the form.
3. Select the button and change its **Text** in the inspector, for example to “Greet”.
4. Double-click the button: the `Button1_click` method is created and the code opens. Type `self.Label1.config(text="Hello, " + self.Entry1.get())`.
5. Press **F9**. The window opens in its own system window.

Chapter 3

The Designer

Projects, components, inspector, events, generated code and running.

Projects

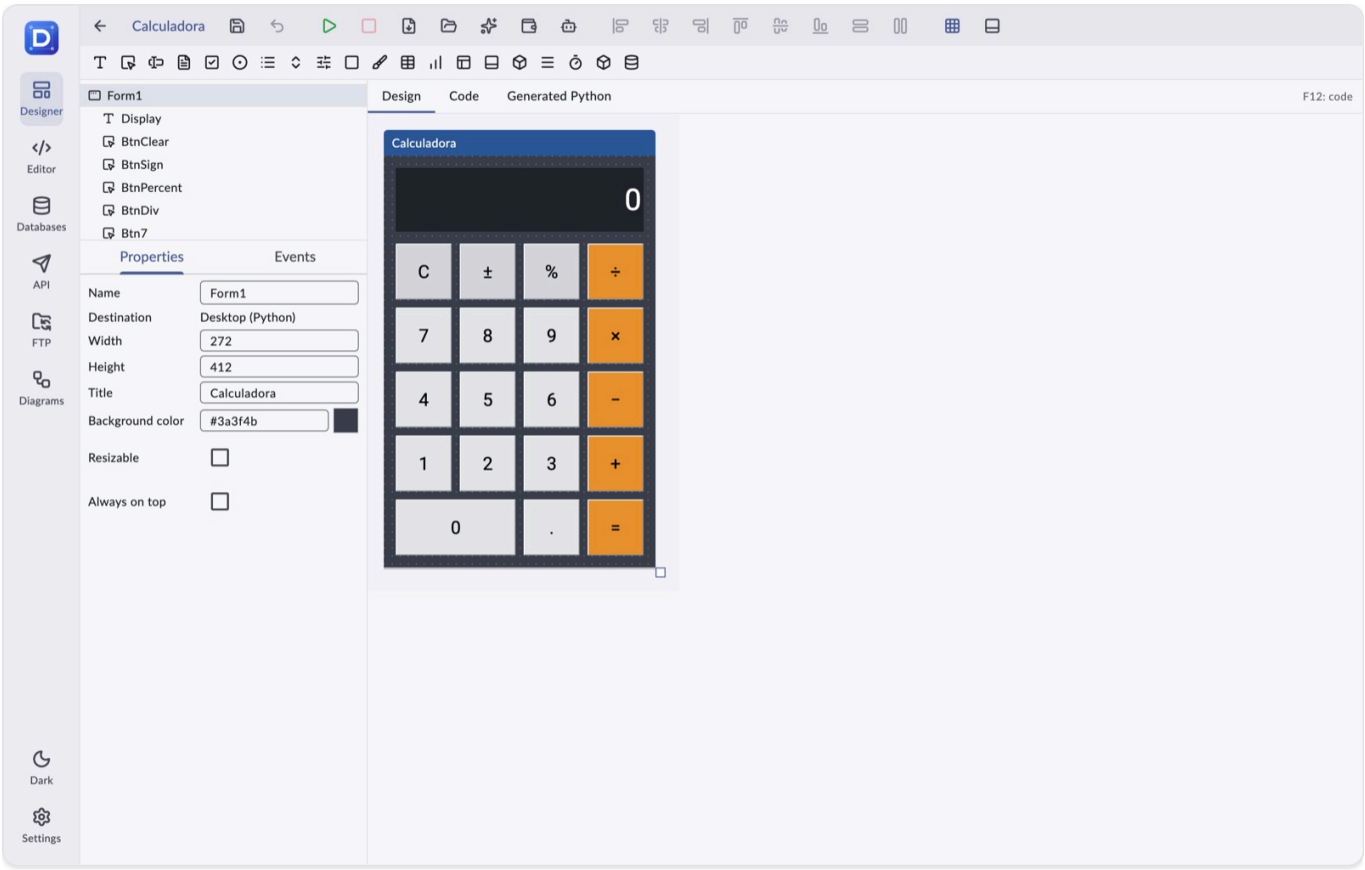
The project list shows the workspace projects and those opened from other folders. At the top you see the Python found (Python and Tk versions) and the button to **Look for Python again**.

- **New project**: asks for the name and the target, **Desktop (Python)** or **Web (Flask)**. The target cannot be changed later.
- **Open folder…**: opens a project stored in any other folder; it stays on the list as a recent one.
- **Workspace**: **Show** opens it in Finder or Explorer and **Change…** picks another folder.
- On each card: **Run**, **Show the folder** and, under **More actions**, **Duplicate** and **Delete** (moves the project to the workspace trash).

> The first time, the studio copies some example projects (calculator, unit converter, tasks, stopwatch, customer form, Odoo contacts…), in desktop and web versions. They are good starting points.

The Designer screen

- **Toolbar**: back to the projects, application name (click to rename it), **Save**, **Undo**, **Run (F9)** in green, **Stop** in red, **Open in browser** (web), **Show the project folder**, **Export the .py file**, **Ask the AI**, **Chat with Claude Code**, alignment, **Snap to grid** and **Show output**.
- **Palette**: a bar of icons; hovering shows the component's name. Drag it onto the form or add it with a double click.
- **Tree**: the window and its components, with those inside a frame or a page indented.
- **Inspector**: **Properties** and **Events** of the selected component or, if none, of the window.
- **Center**: **Design**, **Code** and **Generated Python** tabs. **F12** switches between design and code.



The example calculator in the Designer.

Components

Every component has a **Name**, a position and a size. Almost all accept **Background color**, **Text color** and font (**Font**, **Font size**, **Bold**, **Italic**), and the mouse events (**Click**, **Double click**, **Right click**, **Mouse enter**, **Mouse leave**). Typing components add the keyboard ones (**Focus in**, **Focus out**, **Key down**, **Key up**, **Enter key**).

Component	Own properties	Main events
Label (`Label`)	Text, Alignment, Justification, Relief, Line wrap	Mouse
Button (`Button`)	Text, State, Relief	**On press** (`Button1_click`)
Text field (`Entry`)	Initial text, Justification, Mask with (e.g. `*` for passwords), State, Linked table and Linked field	Keyboard and mouse
Password (`Password`)	Placeholder text, Button to show the password, Maximum length, State	Keyboard and mouse
Image (`Image`)	File (in the project's `images` folder), Fit (original size, whole without distortion, fill and crop, stretch), Border	Mouse

Component	Own properties	Main events
Long text (`Text`)	Initial text (several lines), Line wrap, State	Keyboard and mouse
Checkbox (`Checkbutton`)	Text, State, Checked	**On press**
Option (`Radiobutton`)	Text, State, Group, Value, Checked	**On press**
List (`Listbox`)	Items (one per line), Selection (browse, single, multiple, extended)	**On select**
Drop-down list (`Combobox`)	Items (one per line), Initial text, Justification, State (normal, readonly, disabled)	**On select**
Slider (`Scale`)	Title, From, Until, Step, Value, Orientation, Show value	**On change** (receives the value)
Frame (`Frame`)	Relief, Border. Holds other components	Mouse
Canvas (`Canvas`)	Background color. For drawing from code	Mouse, **Drag**, **Button release**, **Mouse move**
Data table (`Treeview`)	Data source, SQL query, Columns (field|title|width), Limit, Sequence, Load on open	**On select**
Progress bar (`Progressbar`)	Value, Maximum, Mode (determinate or indeterminate), Orientation	Mouse
Page control (`PageControl`)	Pages (one per line), Active page. Each page holds its own components	**On page change**
Status bar (`StatusBar`)	Text. Always takes the bottom edge of the window	Mouse
Ribbon (`Ribbon`)	Tabs, groups and buttons (indented; “Button | icon”), Active tab	**On button press**, **On tab change**
Menu (`Menu`)	Options (one per line; indent = submenu; `-` = separator). Non-visual: shown at the top of the window	**On choosing an option** (receives the option)
Timer (`Timer`)	Interval (ms), Active. Non-visual	**Every interval**
API integration (`RestClient`)	Base URL, Authentication (none, Bearer token, basic or a key in a header; the secret, preferably in an environment variable), Headers, Timeout. Non-visual	**When a call finishes**, **If a call fails**

Component	Own properties	Main events
Tray icon (`TrayIcon`)	Hover text, Image, Menu (one option per line). Non-visual	**On icon click**, **On choosing a menu option**
Open dialog (`OpenDialog`)	Title, Types ("Text|*.txt"), Default extension, Initial folder, Choose several files. Non-visual	**On choosing a file**
Save dialog (`SaveDialog`)	Title, Types, Default extension, Initial folder, File name, Ask before overwriting. Non-visual	**On choosing a file**
Data connection (`Connection`)	Connection (one of those saved in **Databases**). Non-visual	—

Menu, Progress bar, Status bar, Timer and Page control arrive with version 1.0.5; Password, Image, Ribbon, API integration, Tray icon and the Open and Save dialogs, with 1.0.6. In a web app, components without an equivalent say so in the inspector.

Moving, aligning and editing

- Click selects; dragging moves; the handles resize. With **Snap to grid** on, everything fits the grid.
- The arrow keys move the selection one grid step; with Alt (Option), one pixel; with Shift, they resize.
- With two or more components selected, **Align left**, **Center horizontally**, **Align right**, **Align top**, **Center vertically**, **Align bottom**, **Same width** and **Same height** become available (relative to the first one selected).
- Right-click offers **Cut**, **Copy**, **Paste**, **Delete**, **Bring to front** and **Send to back**. What you copy can be pasted in another window, children included.
- **Undo** (⌘Z / Ctrl+Z) reverts each design change. If the code was edited by hand after a step, that step restores the design but not the code, and you are told so.

Events and code

The window's code is a Python class named after the form that inherits from `FormDesign`, the class the studio generates. `self.root` is the window and each component is `self.<Name>`.

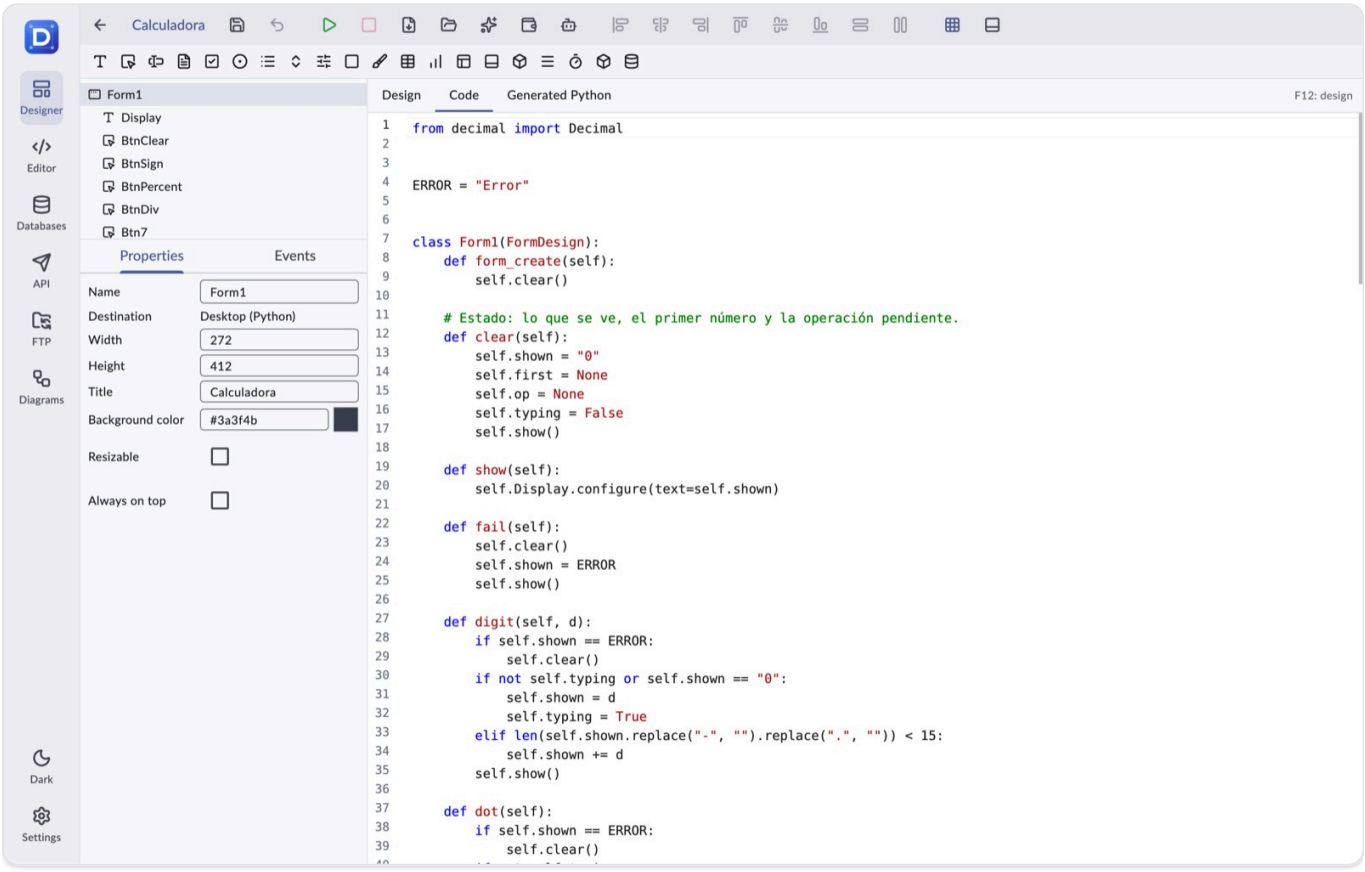
- Double-clicking a component opens its main event (`Button1_click`, `Listbox1_select`…) and creates it if needed.
- On the inspector's **Events** tab you type or choose the method for each event; **Go to code** jumps to it.
- Renaming a component renames its default methods and the references in the code.

- The window has its own events: **On open** (`form_create`), **On close**, **Key down** and **On resize**, and its properties **Title**, **Background color**, **Resizable** and **Always on top**.

```
class Form1(FormDesign):
    def form_create(self):
        self.total = 0
        self.StatusBar1.config(text="Ready")

    def Button1_click(self):
        self.total += 1
        self.Label1.config(text=f"Clicked {self.total} times")
```

> ⚠ Put what should happen at startup in the **On open** event (`form_create`). If you define your own `__init__`, call `super().__init__()` first.

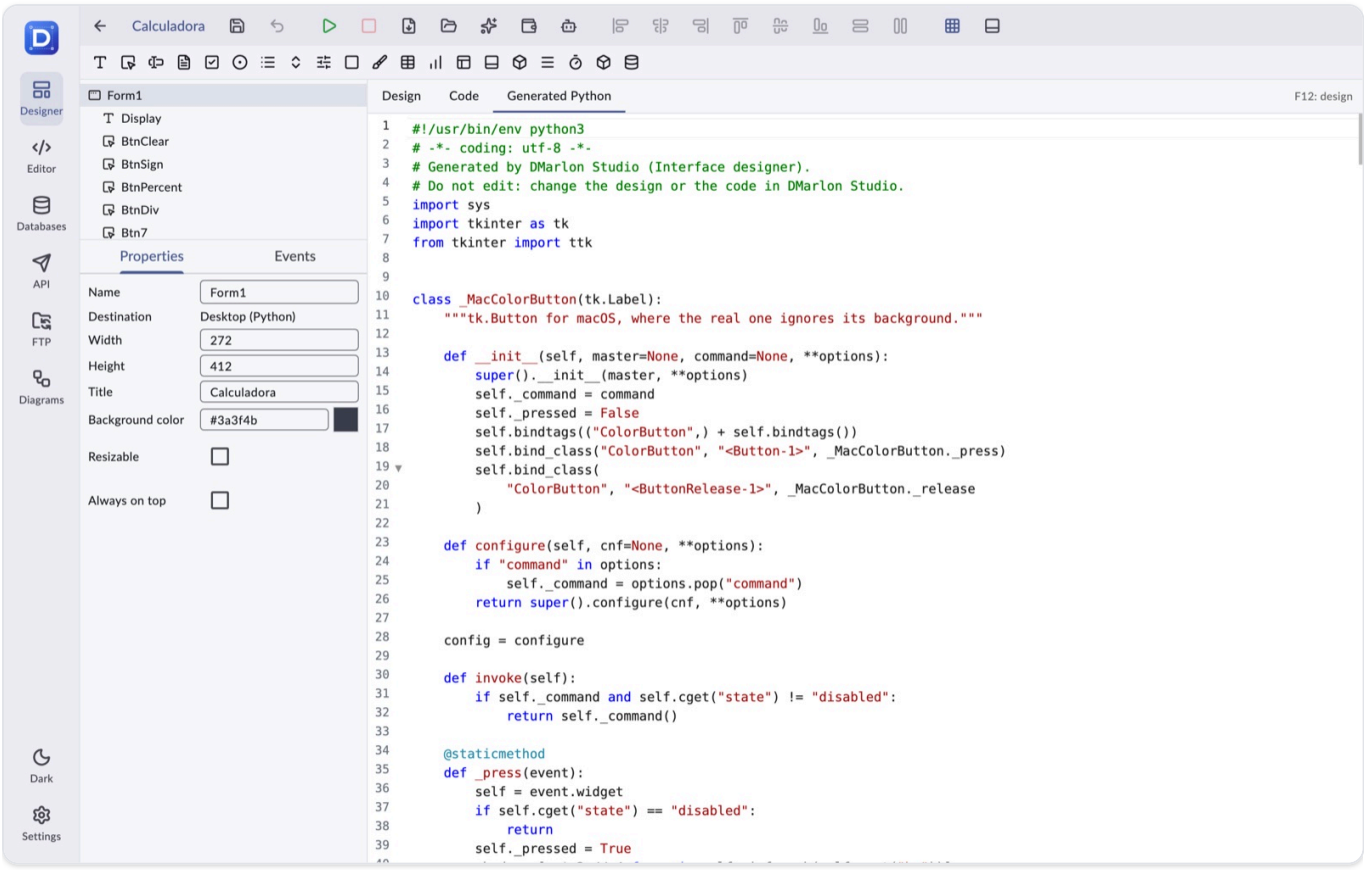


The Code tab: the window class with its handlers.

Generated Python and export

The **Generated Python** tab shows, read-only, the complete file that runs: the design turned into code plus your class. When you run, it is written to `build/<project>.py` inside the project folder.

Export the .py file saves that file wherever you want. It works with any Python 3 with tkinter (or Flask, for web apps) without the studio.



The generated Python: not edited, rebuilt from the design and the code.

Run, stop and see the output

- **Run (F9)** saves the project and launches the window in its own system window, with your user's permissions.
- **Stop** closes the running window or server.
- **Show output** opens the panel with what the app writes (`print`, Python errors) and the Python used.
- For a web app, the studio starts the Flask server on this computer and opens the browser; **Open in browser** opens it again. It can only be reached from this computer.

Data connections

1. Save the database in the **Databases** module (SQLite, PostgreSQL or MySQL/MariaDB).
2. Add a **Data connection** to the form and choose that connection in its **Connection** property.
3. Add a **Data table**, choose the connection in **Data source** and write the **SQL query** and the **Columns** (`field|title|width`).
4. To edit the selected row, link a **Text field** with **Linked table** and **Linked field**.

From code, `self.Connection1.query(sql, params)` returns rows and `self.Connection1.execute(sql, params)` changes data (always with `?` parameters, never by joining text). `self.Treeview1_refresh()` reloads the table.

> Passwords are never written to the generated file: they reach the process in `DMSTUDIO_DB_…` environment variables. An exported file reads them from the environment.

Chapter 4

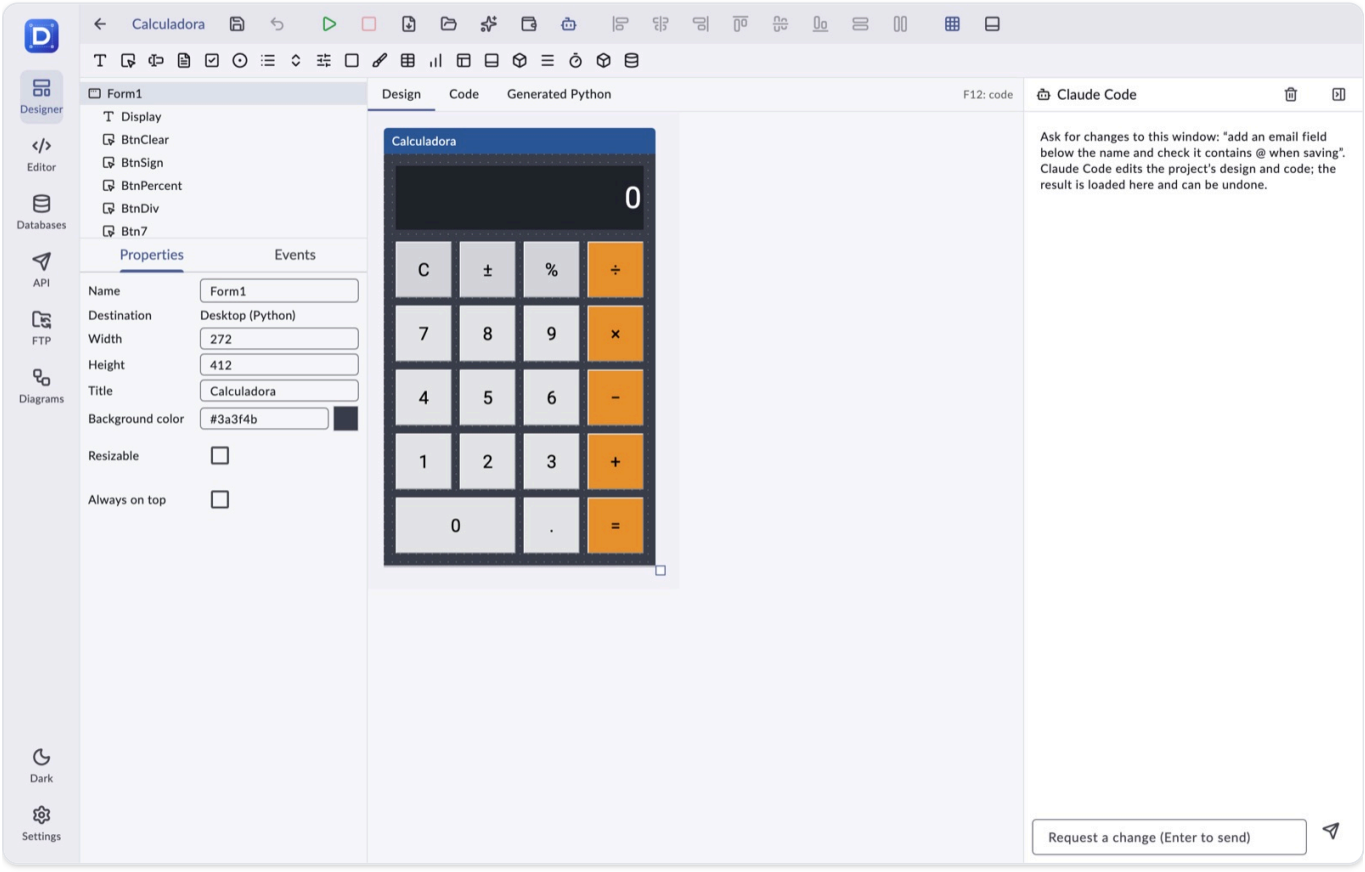
AI in the Designer

Chat with Claude Code and “Ask the AI”.

Chat with Claude Code

The **Chat with Claude Code** button (the robot on the toolbar) opens a panel on the right. Describe the change and press Enter (Shift+Enter adds a new line).

- It uses the Claude Code installed on your computer and your session: no API key needed.
- Before each message the studio saves the project. Claude Code reads and edits the design (`project.dmsproj`) and the window's code, without a terminal.
- When it finishes, the new design and code are loaded into the Designer as a single step that can be undone.
- The conversation continues between messages; the trash button starts a new one.



The Claude Code panel, to the right of the form.

Ask the AI

Ask the AI uses the provider set in **Settings** (Claude by default) and has three modes:

- **Switch** changes the current window (“Add a phone field and align the buttons”).
- **New window** designs a window from scratch; you can **Attach a sketch** (an image).
- **Code of** a component: with a component selected, it writes the method for its main event.

The answer arrives as an **AI proposal**: you review it before applying it and, once applied, it is undone in one step.

Chapter 5

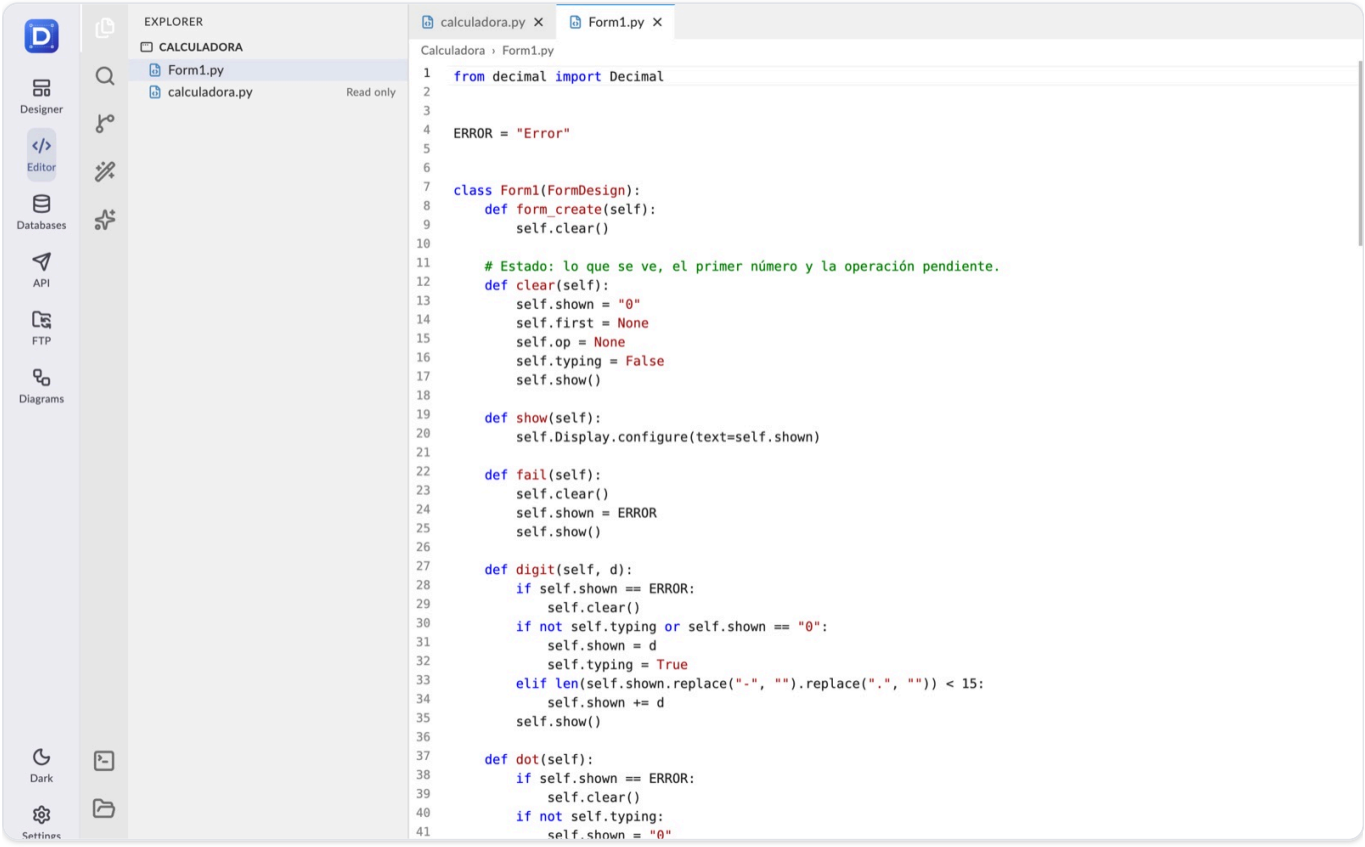
The Editor

Code editor with explorer, search, terminal, git and AI.

Opening a folder

With no folder open you get **Open folder...**, **Open DMarlon Studio** (the studio's documents folder), **Clone repository...** and the list of **Recent** folders.

If a project is open in the Designer, the Editor shows its code and its generated Python as **Project open in the Interface designer**: what you type here is the same text as in the Code tab.



The Editor: activity bar, explorer and tabs.

Explorer and tabs

- The activity bar holds **Explorer**, **Search**, **Source control**, **Claude: implement** and **AI chat**, with the terminal and **Open folder** at the bottom.
- The explorer offers **New file**, **New folder**, **Rename**, **Delete**, **Copy path**, **Open in terminal** and **Collapse all**.

- Tabs have syntax highlighting for dozens of languages and line numbers. ⌘S / Ctrl+S saves; closing something unsaved asks first.
- Anything unsaved is kept when you switch modules.

Find and replace

- **Find in file** with **Replace** and **Replace all**.
- The **Search** panel goes through every file in the folder, with **Match case** and **Regular expression**; a click opens the file at that line.

Terminal

⌘J / Ctrl+J shows or hides the terminal, which runs commands in the open folder (`git status`, `python3 app.py`, `npm install` …). The up and down arrows go through the history and **Stop** ends the running command.

Source control (git)

- **Changes**: modified files with their diff; type the **Commit message** and use **Commit all** or **Commit and push**. Also **Amend last commit** and **Undo last commit**.
- **Push**, **Pull** and **Check the remote for changes (fetch)**, with what is pending each way.
- **History** of the branch or of **All branches**: view the commit, **Revert this commit**, **Apply to the current branch (cherry-pick)**, create a branch or a tag from it and **Reset the branch to this commit…**.
- Local and remote **Branches**: create, switch, **Merge**, rename and delete (also on the remote).
- **Stash changes** to switch branches and bring them back later.
- **Remotes…**, **Initialize repository**, **Clone repository…** and **Publish to GitHub…** (public or private) with your **GitHub account** (a token with the “repo” scope).

Claude: implement

Claude: implement asks Claude for a change through the Anthropic API (Claude Opus 5.5 by default). Claude explores the open folder (lists, reads and searches files) and proposes changes to one or more files.

- Nothing is written until you apply it: the **Proposed changes** list shows each file with its added and removed lines; a click opens the diff with **Apply** and **Discard**.

- **Apply all** writes every change; if a file changed on disk while Claude was working, you are warned instead of it being overwritten.
- It needs the Anthropic key in **Settings → Artificial intelligence**.

AI chat

The **AI chat** answers about the open file with the provider from **Settings**. If it proposes the whole file, **Apply to the file** puts it in the tab (unsaved) and **Copy the code** copies it.

Chapter 6

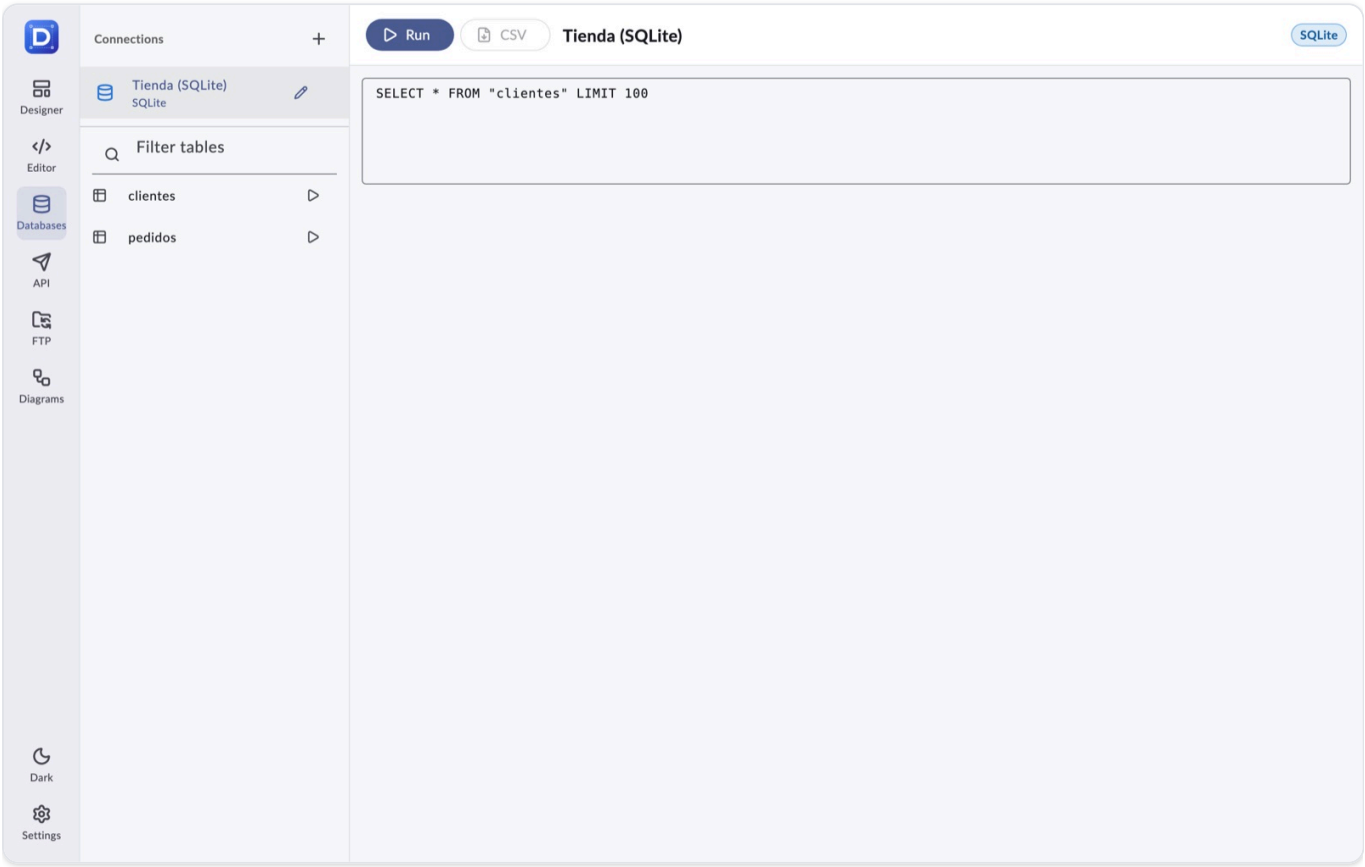
Databases

Connections, tables, SQL queries and CSV export.

Connections

+ adds a SQLite, PostgreSQL, MySQL / MariaDB or MongoDB connection, with server, port, database, user, password and **Encrypted connection (SSL)** depending on the type. **Test connection** checks it can be reached.

The connections saved here are the ones the Designer's windows can use through the **Data connection** component.



A SQLite database with its tables and the query editor.

Tables and queries

- The table list can be narrowed with **Filter tables**; each table opens a sample query.
- **Run** executes the query in the editor and shows the result in a grid.
- Anything that changes data asks for confirmation first.

- **CSV** saves the full result to a file.

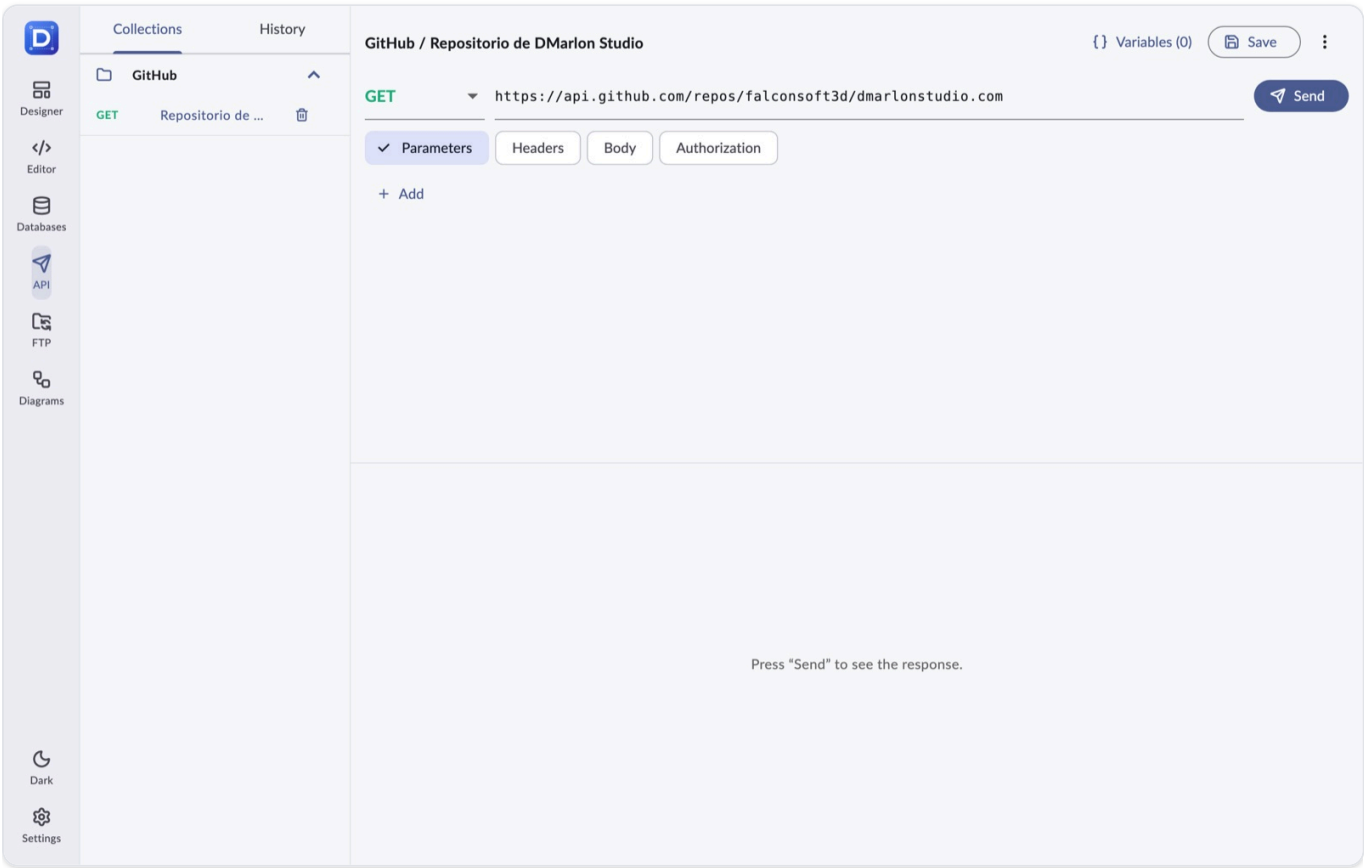
Chapter 7

API client

HTTP requests with collections, variables and history.

Requests

- Pick the method (GET, POST, PUT, PATCH, DELETE…), type the URL and click **Send**.
- **Parameters**, **Headers**, **Body** (JSON, form or text) and **Authorization** (none, Bearer or Basic) tabs.
- The response shows the status code, the time, the headers and the body.
- **Copy as curl** gets the request ready for the terminal.



A request saved in the GitHub collection.

Collections, variables and history

- **Save** stores the request in a **Collection**; **Save as new…** makes a copy.
- **Environment variables…** defines values such as `base_url` or `token`, used in the URL, headers and body as `{{base_url}}`.
- The **History** keeps the last 100 requests sent; **Clear history** empties it.

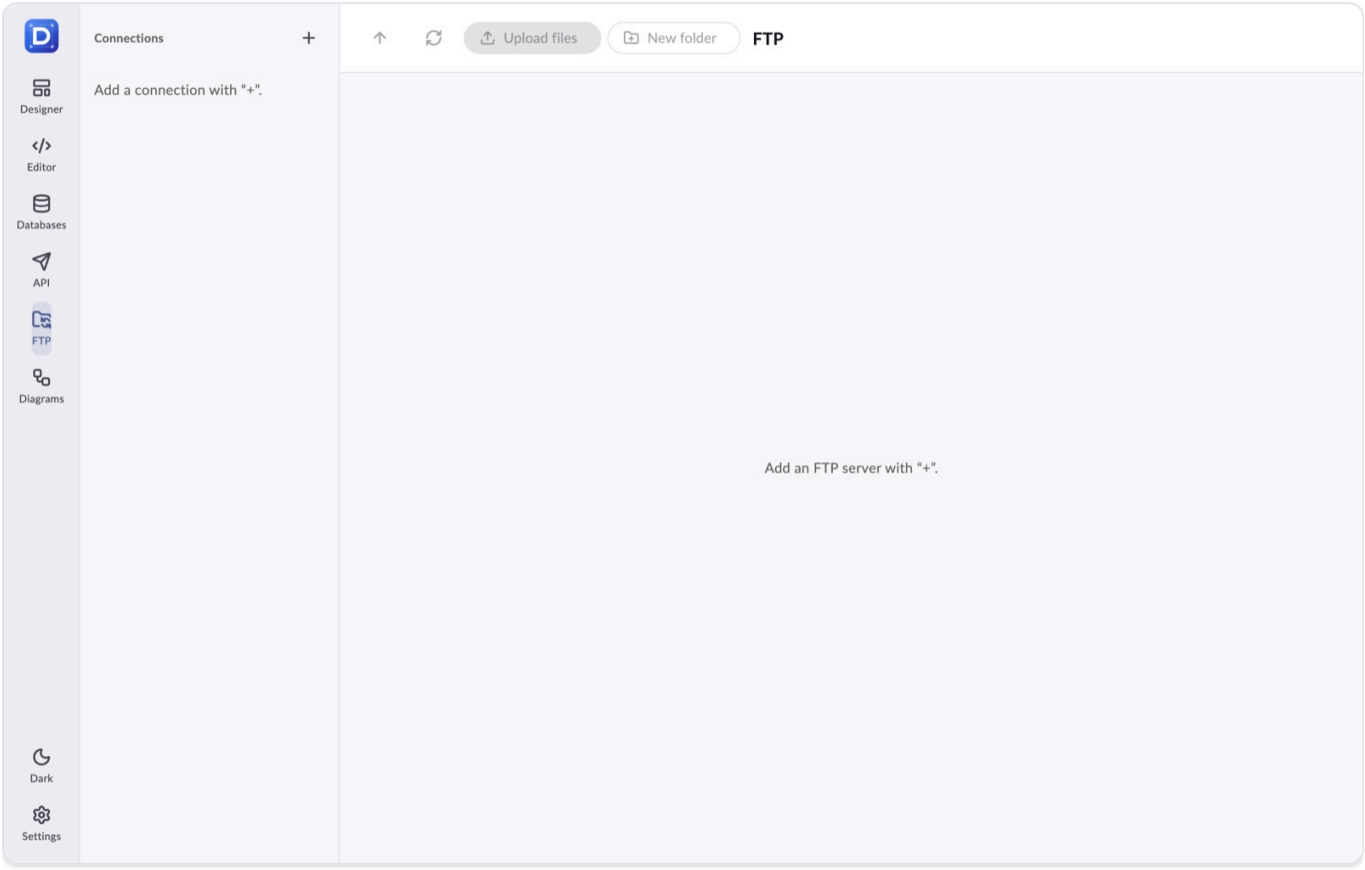
Chapter 8

FTP

FTP, FTPES and FTPS servers.

Servers and files

- **+** adds a server with its **Security**: **FTP (unencrypted)**, **FTPES (explicit TLS)** or **FTPS (implicit TLS)**.
- Choose a server to browse its folders; **Parent folder** goes up a level.
- **Upload files**, **Download**, **New folder**, **Rename…** and **Delete** (folders, with their contents).



The FTP client, ready to add a server.

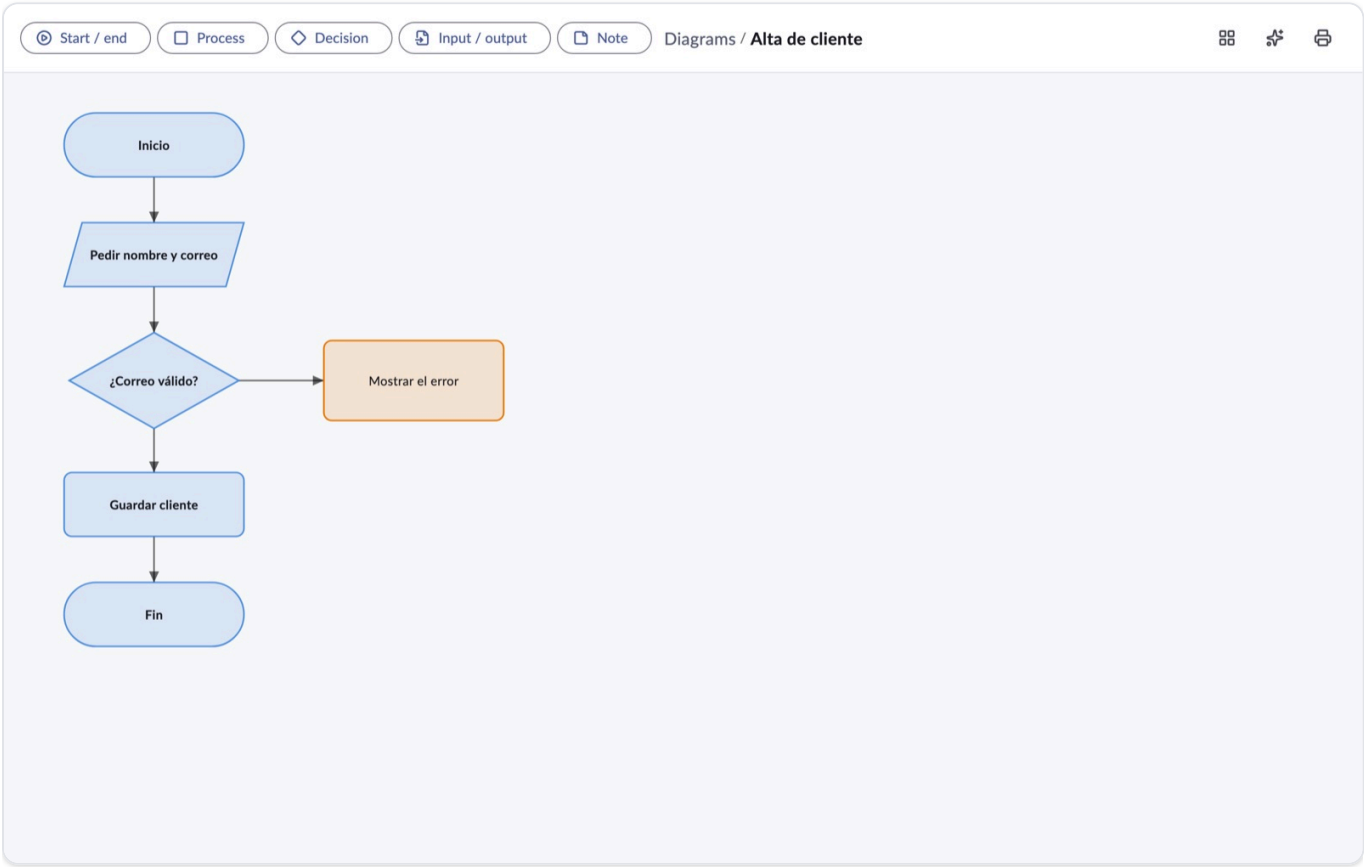
Chapter 9

Diagrams

Flowcharts and database diagrams, with AI and PDF.

Flowcharts and database diagrams

- **Flowchart**: **Start / end**, **Process**, **Decision**, **Input / output** and **Note** shapes. Press → next to a shape to connect it to another; connections can have a label (e.g. “Yes”).
- **Database**: tables with their columns, **Primary key** and foreign keys, and the relations between tables. **Diagram SQL** shows the SQL that creates it.
- **From a database** draws the diagram of a saved connection (optionally, only the tables starting with a prefix).
- **With AI** creates the diagram from a description and **Change with AI** modifies it.
- **Arrange** lays the shapes out again and **Export to PDF** saves it for printing or sharing.



A flowchart for registering a customer.

Chapter 10

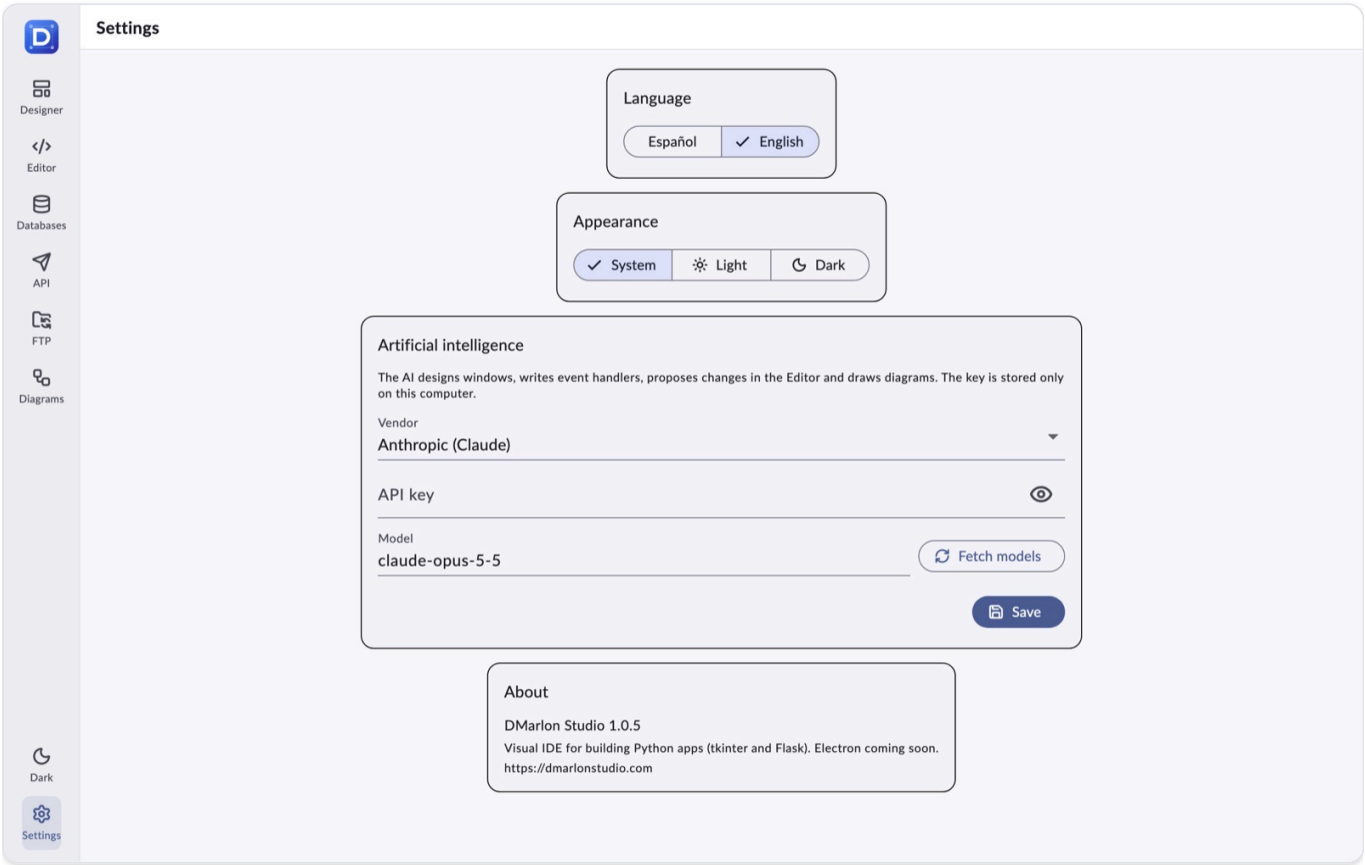
Settings

Language, appearance and artificial intelligence.

Artificial intelligence

Choose the provider (**Vendor**): Anthropic (Claude), OpenAI, Gemini, xAI, DeepSeek, Mistral, Groq, Cohere, OpenRouter, Perplexity, Ollama or a compatible custom API. Type the **API key** and the **Model**; **Fetch models** lists those on your account. Claude (`claude-opus-5-5`) is the default provider.

The key is stored on this computer only. Each provider remembers its key and model when you switch between them.



Settings: language, appearance, artificial intelligence and version.

Language, appearance and version

- **Language**: Español or English, instantly.
- **Appearance**: System, Light or Dark.
- **About** shows the installed version.

Chapter 11

Keyboard shortcuts

Every Designer and Editor shortcut.

Designer

Shortcut	Action
F9	Run
F12	Switch between design and code
⌘S / Ctrl+S	Save the project
⌘Z / Ctrl+Z	Undo
⌘C, ⌘X, ⌘V (Ctrl on Windows)	Copy, cut and paste components
Delete / Backspace	Delete the selection
Arrows	Move one grid step (with Alt, one pixel)
Shift + arrows	Resize
Double-click on the palette	Add the component
Double-click on a component	Open its main event
Enter in a chat	Send (Shift+Enter: new line)

Editor

Shortcut	Action
⌘S / Ctrl+S	Save the file
⌘J / Ctrl+J	Show or hide the terminal
↑ / ↓ in the terminal	Go through the command history
Enter in a chat	Send (Shift+Enter: new line)

Chapter 12

Troubleshooting

The most common messages and how to solve them.

Python not found, or no tkinter

- **Python was not found**: install it (python.org or Homebrew) and click **Look for Python again** on the project list.
- **Found … but it does not include tkinter**: on macOS, `brew install python-tk`; on Linux, `sudo apt install python3-tk`; on Windows, reinstall Python ticking **tcl/tk and IDLE**.
- **Tk 8.5 is old and does not honor the colors of the design**: that is the Python bundled with macOS. Install one from python.org or Homebrew.

Flask is missing

Web apps need Flask in the same Python. The **Flask is missing** message shows the exact command, for example `python3 -m pip install flask`.

The app won't open (unsigned)

- **macOS**: the first time, right-click DMarlon Studio → **Open** → **Open**.
- **Windows**: on “Windows protected your PC”, **More info** → **Run anyway**.

AttributeError: 'Form1' object has no attribute 'root'

This happened when the window's code defined its own `__init__` without calling `super().__init__()`: that method replaced the one that creates the window. Since version 1.0.5 the window and its components are created before your `__init__`, and the **On open** event runs after it.

```
class Form1(FormDesign):
    def __init__(self):
        super().__init__()   # first
        self.current = "0"

    # Even better: the On open event
    def form_create(self):
        self.current = "0"
```

Claude Code was not found

Install Claude Code from claude.com/claude-code and run `claude` once in a terminal to sign in. The studio looks for it on the PATH and in `~/.local/bin`.

Chapter 13

Roadmap

What comes next.

More targets

The same designer will generate and compile apps for more environments. You will choose the target when creating a project:

Target	Status
Python + tkinter (desktop)	Available
Python + Flask (web)	Available
Electron + React	On the way
C++ with FLTK	On the way
Pascal with Lazarus (LCL)	On the way

Richer projects

- Several forms per project, with a **Project** node in the tree.
- The application icon, set in the project.
- Building executables (PyInstaller for Python).